

22 popquiz Faradays law

⚠ This is a preview of the published version of the quiz

Started: Apr 3 at 6:15am

Quiz Instructions

No, you don't = Lenz law or "Carmen law" -



Question 1 20 pts

Slide 7

- 1) A flat coil is in a uniform magnetic field. The magnetic flux through the coil is greatest when the plane of its area is
- A) at 45° with the magnetic field.
 - B) parallel to the magnetic field.
 - C) perpendicular to the magnetic field.

☐

C

☐

A

☐

B



D



Question 2 20 pts

A coil of wire containing N turns is in an external magnetic field that is perpendicular to the plane of the coil and it steadily changing. Under these circumstances, an emf ε is induced in the coil. If the rate of change of the magnetic field and the number of turns in the coil are now doubled (but nothing else changes), what will be the induced emf in the coil?

A) 2ε B) ε C) 4ε D) $\varepsilon/4$ E) $\varepsilon/2$ 

C



A



B



D



Question 3 20 pts

A closed flat loop conductor with radius 2.0 m is located in a changing uniform magnetic field. If the emf induced in the loop is 2.0 V, what is the rate at which the magnetic field strength is changing if the magnetic field is oriented perpendicular to the plane in which the loop lies?

A) 1.0 T/s

B) 0.16 T/s

C) 0.080 T/s

D) 2.0 T/s



B



A



C



D



Question 4 20 pts

Tricky question. Refer to slide 7. Don't rush. understand the geometry. could be TMI

A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are parallel.

A) $0 \text{ T} \cdot \text{m}^2$

B) $3.1 \times 10^{-3} \text{ T} \cdot \text{m}^2$

C) $5.5 \times 10^{-3} \text{ T} \cdot \text{m}^2$

D) $6.3 \times 10^{-3} \text{ T} \cdot \text{m}^2$

☐

A

☐

B

☐

C

☐

D

☒

Question 5 20 pts

Slide 7. TRicky question. In the equation for flux, you want the angle between B and the vector perpendicular to the plane of the surface A. YOU DON'T WANT the angle between the plane and B !

so are you sure its 30?

A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are at an angle of 30° .

A) $0 \text{ T} \cdot \text{m}^2$

B) $3.1 \times 10^{-3} \text{ T} \cdot \text{m}^2$

C) $5.5 \times 10^{-3} \text{ T} \cdot \text{m}^2$

D) $6.3 \times 10^{-3} \text{ T} \cdot \text{m}^2$

☐

B

☐

A

☐

D

☐

C

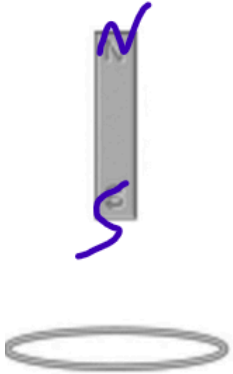
☒

Question 6 10 pts

Bonus

Tricky. The loop is trying to move away from the magnet. So the magnet says "no, you don't". So its trying to stop the loop from moving away. I call the "Carmen" law.

A bar magnet is oriented above a copper ring, as shown in the figure. If the magnet is kept fixed while the ring is dropped, what is the direction of the current induced in the ring, as viewed from *below* the ring?



- A) There is no current in the ring.
- B) counterclockwise
- C) clockwise

☐

C

☐

A

☐

B

☐

D

Quiz saved at 6:15am

Submit Quiz